Chapter 3. Relativistic Mechanics

3.1 Mass-Energy Equivalence

The most famous equation in modern physics defines the equivalence between mass and energy. The relation was first derived by Einstein in his 1905 paper and reads as follows:

$$E^2 = p^2c^2 + (mc^2)^2 \tag{3.1}$$

where p is the relativistic momentum of the particle, c is the speed of light in vacuum, and m is the invariant rest mass. This continuity equation must be preserved under any Lorentz transformation.

3.2 The Schrödinger Equation

In quantum mechanics, the time-dependent Schrödinger equation is:

$$i\hbar \, \partial\psi/\partial t = \hat{H}\psi \tag{3.2}$$

A canonical integral that appears throughout statistical mechanics:

$$\int_0^\infty e^{-x} \, dx = 1 \tag{3.3}$$

The exponential function admits the Taylor series expansion:

$$e^x = \sum_{n=0}^{\infty} x^n/n! \tag{3.4}$$

3.3 Notation in Chemistry

Subscripts are equally important in chemical formulae. Water is written as H_2O, sulfuric acid as H_2SO_4, and Avogadro's number is $N_A = 6.022 \times 10^{23}$ mol^{-1}.

3.4 Greek Symbols in Physics

Common Greek letters: α (alpha), β (beta), γ (gamma), Δ (Delta), λ (lambda), μ (mu), π (pi), σ (sigma), φ (phi), ω (omega). Operators: ∇ (nabla), ∂ (partial), ∫ (integral), √ (radical).